

PayFlow GST - System Design Document (HLD + LLD)

Product: PayFlow GST
Frontend: React
Backend: Node.js (Express)
Database: PostgreSQL
ORM: Prisma
Authentication: JWT
Storage: AWS S3
Deployment: Docker + AWS
Architecture Style: Multi-tenant SaaS

1. System Overview

PayFlow GST is a multi-tenant SaaS platform that enables Indian SMEs to create GST invoices, track receivables, manage collections, and generate compliance reports.

Primary Modules:

- Authentication
- Business Management
- Customer Management
- Invoice Management
- Payment Tracking
- Collections & Reminders
- Reporting & Analytics

2. High Level Architecture

React SPA
|
API Gateway / Nginx
|
Node.js Express API
|

Service Layer

|

Prisma ORM

|

PostgreSQL

External Services:

- AWS S3
- Email Service
- WhatsApp Service
- Razorpay

3. Frontend Architecture

Framework: React

Libraries:

- React Router
- TanStack Query
- Axios
- React Hook Form
- Zod
- Material UI

Pages:

- Login
- Registration
- Dashboard
- Customers
- Invoices
- Payments
- Reports
- Settings

State Management:

- Context API
- React Query Cache

4. Backend Architecture

Framework: Express.js

Layers:

1. Routes
2. Controllers
3. Services
4. Repositories
5. Prisma Data Access

Principles:

- Separation of concerns
- Stateless APIs
- Dependency injection ready

5. Authentication Design

JWT Authentication

Flow:

User Login

- > Verify Credentials
- > Generate Access Token
- > Generate Refresh Token
- > Return Tokens

Authorization:

- Owner
- Manager
- Staff

RBAC implemented at middleware layer.

6. Multi-Tenant Design

Every record belongs to a Business.

Tenant Isolation:

business_id included in:

- customers
- invoices
- payments
- users

Every API validates tenant ownership.

7. Invoice Service Design

Responsibilities:

- Create invoice
- Calculate GST
- Generate invoice number
- Generate PDF
- Track status

Status Flow:

Draft

-> Unpaid

-> Partial

-> Paid

Overdue calculated by scheduler.

8. Payment Service Design

Responsibilities:

- Record payment
- Validate amount
- Update invoice status
- Maintain audit trail

Rules:

Total payment cannot exceed invoice total.

9. Reminder Service Design

Background Scheduler

Events:

- Due in 3 days
- Due today
- Overdue 7 days
- Overdue 15 days

Channels:

- Email
- WhatsApp

Queue Processing:

Cron Job

- > Fetch eligible invoices
- > Send reminder
- > Store status

10. Reporting Service Design

Reports:

- GST Summary
- Outstanding Report
- Customer Ledger
- Revenue Report

Export Formats:

- PDF
- Excel

11. Database Access Layer

Prisma ORM

Benefits:

- Type Safety
- Migration Support

- Query Optimization

Repository Pattern:

Controller

-> Service

-> Repository

-> Prisma

12. API Design Standards

REST APIs

Naming:

GET /customers

POST /customers

GET /invoices

POST /invoices

Response Format:

```
{  
  success,  
  data,  
  message  
}
```

13. Security Architecture

Security Controls:

- HTTPS Only
- JWT
- Password Hashing (bcrypt)
- Rate Limiting
- Input Validation
- SQL Injection Protection
- XSS Protection
- CSRF Protection
- Audit Logs

Sensitive Data:
Encrypted at rest where applicable.

14. Logging & Monitoring

Application Logs
Audit Logs
Error Logs

Tools:
- Winston
- Morgan

Monitoring:
- CloudWatch
- Grafana
- Prometheus

15. Caching Strategy

Redis Cache

Use Cases:
- Dashboard Metrics
- Frequently Accessed Reports
- Session Blacklists

16. File Storage Design

AWS S3

Files:
- Invoice PDFs
- Business Logos
- Report Exports

Naming Convention:
business_id/entity/timestamp

17. Deployment Architecture

React
-> Vercel

Node.js API
-> Docker
-> AWS ECS

PostgreSQL
-> AWS RDS

Redis
-> AWS ElastiCache

Files
-> AWS S3

18. CI/CD Pipeline

GitHub

Pipeline:
Code Push
-> Build
-> Test
-> Lint
-> Security Scan
-> Deploy

Environments:
- Development
- Staging
- Production

19. Scalability Strategy

Horizontal Scaling

API:

Multiple containers

Database:

Read replicas

Cache:

Redis

Files:

S3

Queue:

BullMQ

20. Disaster Recovery

Daily Backups

Point-in-Time Recovery

Multi-AZ Database

Infrastructure as Code

Recovery Targets:

RPO: 15 Minutes

RTO: 1 Hour

21. Future Architecture (V2)

- Inventory Module
- Purchase Orders
- AI Collection Assistant
- GST Filing Integration
- Mobile App

- Analytics Warehouse